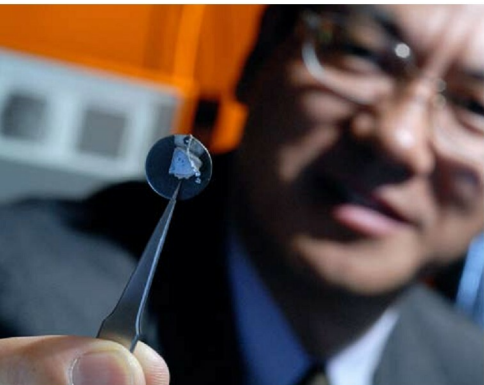


Powering **NANO** Devices

Nanogenerators have the potential to replace conventional batteries powering nanodevices and nanorobots. These self-powered generators can also be implanted in the human body to power the drug delivery system. Read on to know more about their unique advantages



Georgia Tech Professor Zhong Lin Wang holds a sample nanowire array that can be used to power nanometer scale devices (Photo courtesy: Georgia Tech)

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Researchers have been trying to develop new technologies to power nanometer scale devices that won't need bulky energy sources like batteries. Conventional batteries used to power nanosystems in the human body limit their use due to their bulk and toxic contents. The focus of the researchers therefore is on developing nanotechnologies that harvest energy from their environment and self-power nanosystems. The key advantage of nanodevices and nano-

system is that they usually operate at very low power—ranging from nW to μ W—as a result the energy harvested from the body may be sufficient to power the system.

Zhon Lin Wong, a reagents professor at the School of Material Science and Engineering, Georgia Institute of Technology, has presented various options for small-scale energy harvesting by converting the body's mechanical energy (body movements, muscle stretching, blood pressure), vibration energy (acoustic, ultrasonic waves), chemical energy (exothermic reactions), hydraulic energy (flow of

body fluid, blood, contraction of blood vessels), etc, into electrical energy.

These small-scale energy harvesting options have been summed up as following:

Microbial fuel cells. Microorganisms can be used to transfer bioconvertible substrates directly into electricity, where the bacterium acts as electrode and the electrons flow from cathode through a resistor.

Adenosine tri-phosphate energy converter. Biological motor enzymes are fueled by ATP molecules, which consist of Adenine, Ribose and three phosphate groups. As the first phosphate group is removed, a large amount of energy is released. If more amount of energy is needed, the second phosphate group is released. Energy created can be used for chemosynthesis, locomotion (including muscle contraction in animals) and active transport of ions across all membranes. This concept leads to ATP serving as rechargeable batteries inside the human body.

Thermoelectric generator. Thermoelectric generator relies on Seebeck effect (the conversion of temperature differences into electricity), which is about the electric potential observed flowing between two dissimilar metals that form a junction at different temperatures. The voltage produced is proportional to temperature difference between the two ends. An applied temperature difference causes charge carriers in the material, whether they are electrons or holes, to diffuse from the hot side to the cold side. Thermoelectric generator usually has large size because of the requirement of main-